

SEMESTER PROJECT GAME DEVELOPMENT

Design Document

Memory of the Past

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Submitted To:	Sir Behram
Subject:	Game Development
Date of Submission:	06/07/2026
Engine Used:	Unity 6 (6000.5.2f1)
GitHub Repository:	https://github.com/atifkhanfall2024/MemoryPastGameUnity

1. Game Title

Memory of the Past

2. Theme Chosen

Culture and Heritage

The game celebrates elements of Pakistani culture and heritage, including traditional art, clothing, transport, monuments, and festivals.

3. Inspiration

Why this topic was selected:

- Pakistan has a rich and diverse cultural heritage that is often overlooked by younger generations in favor of foreign media and entertainment.
- Many traditional crafts, festivals, and symbols risk being forgotten as digital entertainment increasingly centers on global rather than local content.

What inspired the game idea:

- The classic memory-matching card game format was chosen because it is simple to learn, universally recognizable, and lends itself naturally to teaching players to recognize and associate cultural symbols.
- Instead of generic or foreign imagery, each card was redesigned to represent a distinct piece of Pakistani culture, turning a familiar game mechanic into a small cultural showcase.

Message for players:

- To encourage players — especially younger audiences — to recognize, value, and remember symbols of their own cultural heritage through an engaging and interactive medium.

4. Game Concept

Memory of the Past is a 2D single-player memory-matching card game. Sixteen cards (eight matching pairs) are laid face-down on a 4x4 grid. Each pair represents a different element of Pakistani culture and heritage: Ajrak patterns, truck art, the Basant kite, traditional khussa footwear, the auto-rickshaw, Minar-e-Pakistan, qawwali/dholak music, and the Lok Mela folk festival. The player must find all matching pairs before a countdown timer runs out.

5. Core Gameplay

What the player does:

- The player clicks on any face-down card to reveal it, then clicks a second card to try to find its match.

Main mechanics:

- Card flip and reveal system, using a unique index per cultural image.
- Match-checking logic: if both selected cards share the same index, they remain face-up (matched); otherwise, both flip back down after a short delay.

- A live countdown timer displayed on-screen throughout gameplay.
- A main menu (Start / Quit) that gates the actual gameplay until the player is ready to begin.
- Sound effects for matches, mismatches, winning, and losing, plus a continuous looping background melody.

Goals and challenges:

- The player's goal is to match all 8 pairs before the 60-second timer expires.
- The core challenge is memory and speed: recalling the position of previously seen cards while racing against the clock.

6. Win/Lose Conditions

Win Condition:

- The player wins by successfully matching all 8 pairs of cards before the timer reaches zero. A "You Win!" screen is then displayed along with the option to play again.

Lose Condition:

- If the 60-second timer reaches zero before all pairs are matched, the game ends automatically and a "Time's Up! Game Over" screen is displayed, along with the option to try again.

7. Screenshots

Development progress and gameplay screenshots:

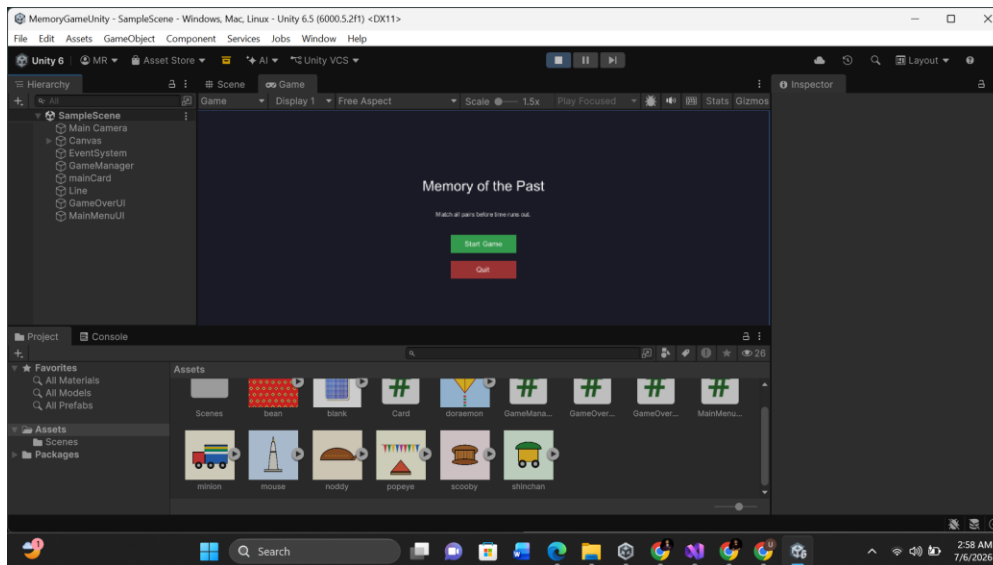


Figure 1: Main Menu screen with game title, description, Start Game and Quit options.

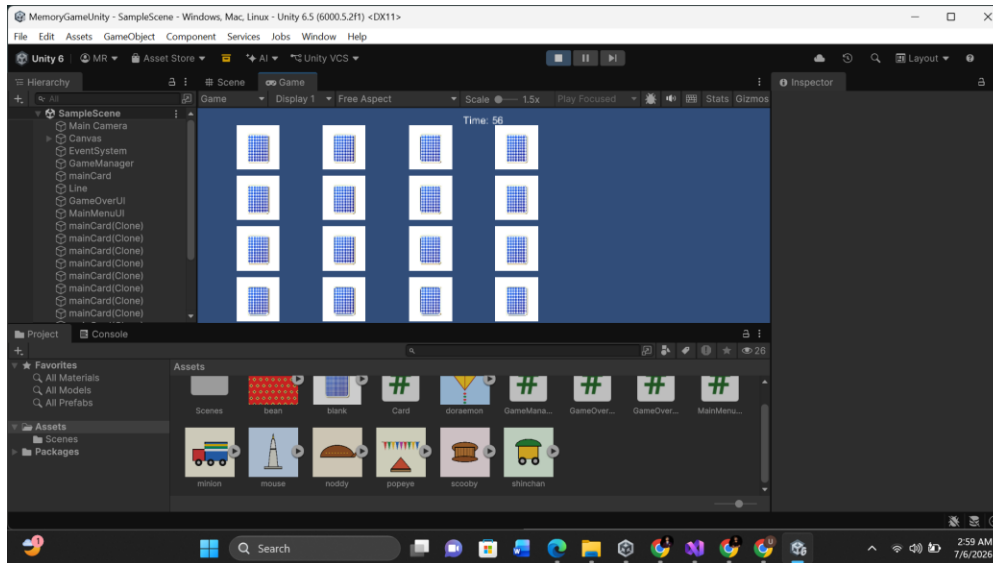


Figure 2: Gameplay scene showing the 4x4 card grid with the live countdown timer.

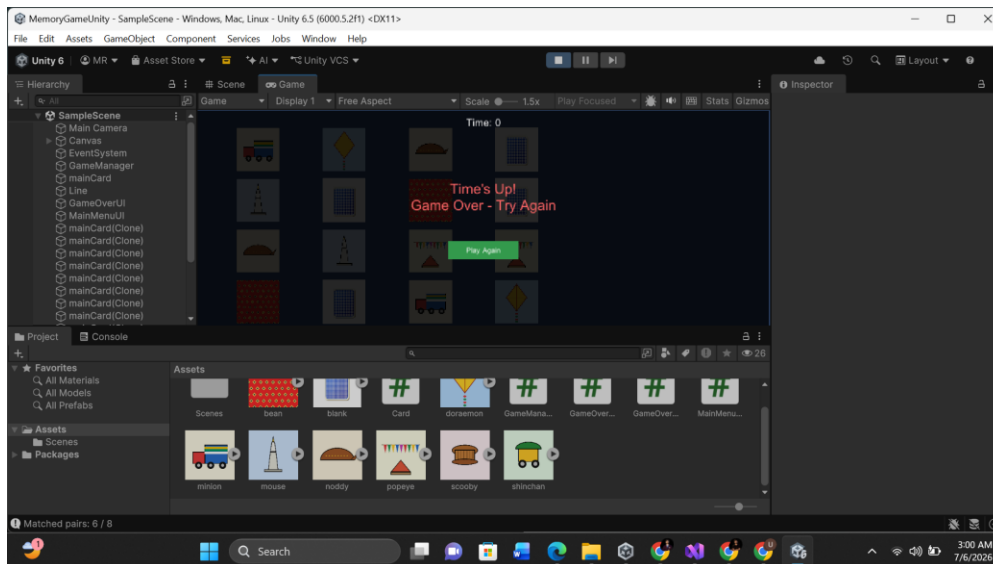


Figure 3: Lose condition triggered — "Time's Up! Game Over" screen with culture-themed cards and Play Again option.

8. Challenges Faced

- Upgrading the base project from an older Unity version (2021 LTS) to Unity 6 without breaking existing functionality.
- The original codebase had an incomplete win-condition system: the match-counting logic existed but was never connected to a visible Game Over screen. This was fixed by adding a pair counter and dynamically locating the Game Over UI component at runtime.
- The project had no lose condition, main menu, or sound. These systems were designed and implemented from scratch: a countdown timer with an automatic loss trigger, a main-menu overlay that gates gameplay until the player presses Start, procedurally generated sound effects, and a looping background melody — all without relying on external asset files.

- Card artwork initially used generic, copyrighted cartoon characters unrelated to the assigned theme. These were replaced with custom-made icons representing Pakistani cultural symbols to properly satisfy the project's theme requirement.
- Newly added artwork initially rendered at the wrong scale on the cards; this was resolved by adjusting the Pixels Per Unit setting on the sprite import settings to match the card prefab's expected size.

9. Requirement Checklist

Self-assessment against the assignment's gameplay requirements:

Requirement	Status
Clear objective	Met
Win and/or lose condition	Met
Main menu	Met
At least one playable level/scene	Met
Basic user interface	Met
Sound effects and/or background music	Met
Game over / completion screen	Met

10. Asset Usage Declaration

This project began from an open-source Unity memory-matching card game template, used in accordance with the course's Asset Usage Policy, which permits open-source assets. The original template's card artwork, game-over system, main menu, sound, background music, and lose condition were incomplete, non-functional, or misaligned with the assigned theme. All of these systems were understood, debugged, rebuilt, or newly implemented as part of this submission, and the author is able to explain each system's implementation in detail.